Project Interim Report On

Hotel Room Reservation Application

(Using Core Java & OOP Concepts)

Object-Oriented Programming With Java

RU-100-01-00012

Submitted By

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Abstract

The Hotel Room Reservation System is a Java-based console application designed to manage hotel room bookings, room availability, and customer billing efficiently. The system helps hotels automate reservation operations and maintain room records in a structured way. This project is developed to demonstrate how real-world hotel reservation systems work using Object-Oriented Programming concepts. Manual room booking management can be time-consuming and error-prone, so this system provides an organized method to handle bookings, cancellations, and billing calculations automatically. The application allows users to book hotel rooms, cancel reservations, check room availability, display room details, and calculate customer bills. The project uses important Java concepts such as classes and objects, encapsulation, interfaces and abstraction, arrays (2D arrays), and constructors and methods. The system simulates a real hotel management environment and improves understanding of object-oriented programming and real-world software development.

Introduction

The Hotel Room Reservation System is a console-based Java application developed using object-oriented programming principles. It is designed to simplify hotel room management operations such as room booking, cancellation, billing, and room status tracking. In real-world hotels, managing reservations manually can create confusion and increase workload. This project automates the reservation process and ensures proper room allocation and billing management. The application uses a Room class to represent hotel rooms, a Hotel class to manage multiple rooms using a 2D array, and a Service interface for billing operations. The project provides a simple and user-friendly system to demonstrate practical implementation of OOP concepts in Java.

Objectives

- To manage hotel room reservations efficiently
- To automate room booking and cancellation operations
- To maintain room availability records
- To calculate customer bills automatically
- To implement interface-based programming
- To apply Object-Oriented Programming concepts
- To understand 2D array handling in Java

Scope

This project is suitable for educational purposes, small-scale hotel management systems, and learning Java programming concepts. It demonstrates understanding of real-world reservation systems. Future enhancements may include database integration, graphical user interface (GUI), online booking system, customer information management, and payment gateway integration.

System Requirements

Hardware Requirements

• Computer/Laptop with minimum 4GB RAM
• Intel i3 Processor or above

Software Requirements

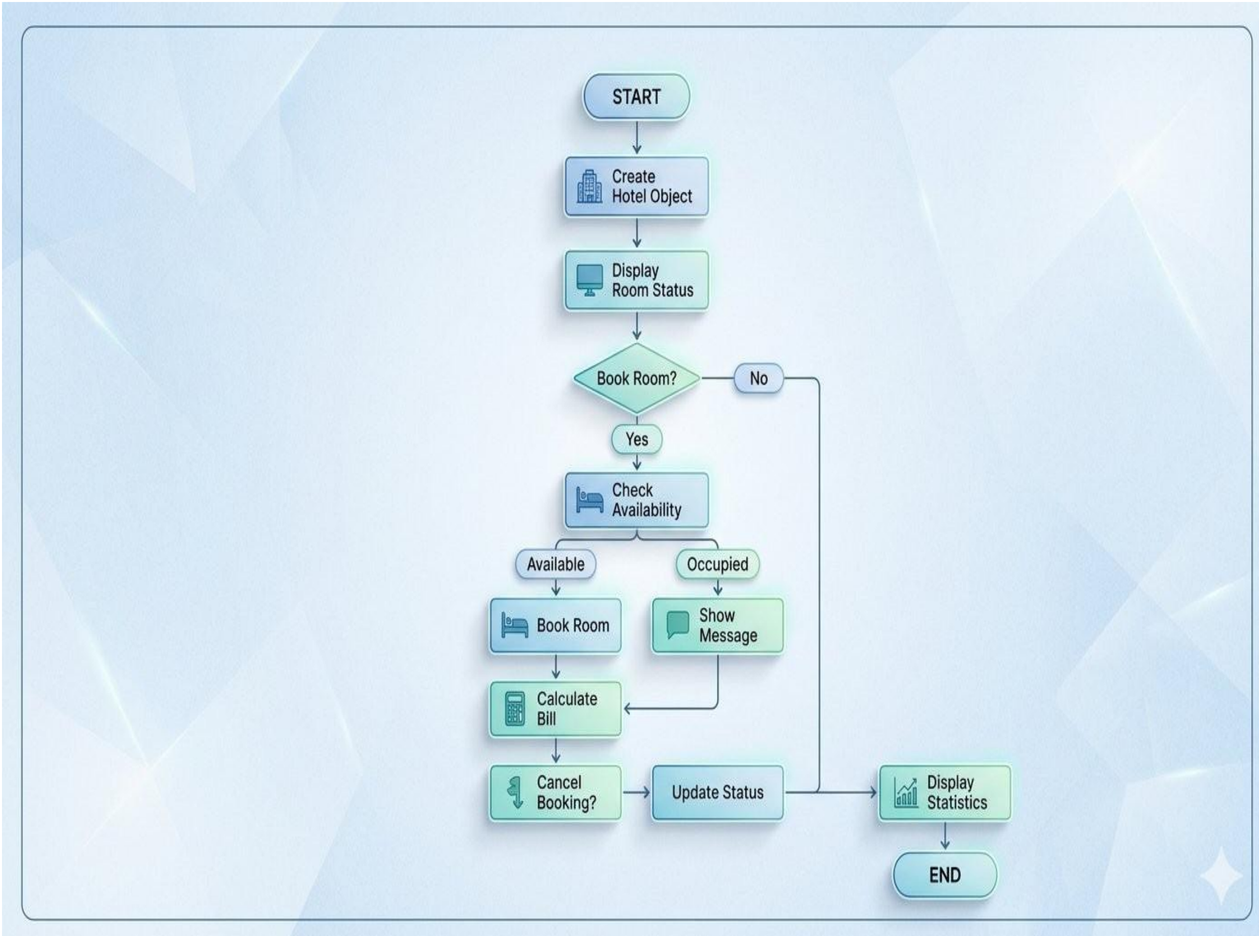
• Java JDK (Java Development Kit)
• IDE: VS Code / Eclipse / IntelliJ IDEA
• Operating System: Windows / Linux / Mac

Methodology

The working of the system follows these steps: (1) The program starts and displays hotel details, (2) The hotel rooms are initialized using a 2D array, (3) The user can book a room by selecting floor and room number, (4) The system checks room availability, (5) If available, the room is booked successfully, (6) The user can cancel existing bookings, (7) Billing is calculated using the Service interface, (8) Room status and hotel statistics are displayed, (9) The system continues until all operations are completed.

Flowchart

The following flowchart illustrates the system flow:


START
Create
Hotel Object
Display
Room Status
Book Room?
No
Yes
Check
Availability
Available
Occupied
Book Room
Show
Message
Calculate
Bill
Cancel
Booking?
Update Status
Display
Statistics
END

Classes, Methods and Variables Used

Class: Room

Variables

• roomNumber
• roomType
• pricePerDay
• isAvailable

Methods

• Room()
• getRoomNumber()
• getRoomType()
• getPricePerDay()
• isAvailable()
• bookRoom()
• cancelBooking()

Interface: Service

• calculateBill(int days, double rate)

Class: Hotel

Variables

• hotelName
• Room[][] rooms
• floors
• roomsPerFloor

Methods

• initializeRooms()
• bookRoom()
• cancelBooking()
• displayAllRoomStatus()

- displayStatistics()
- displayBill()
- getRoom()
- isValidRoom()
- calculateBill()

Class: HotelReservationApp

- main()

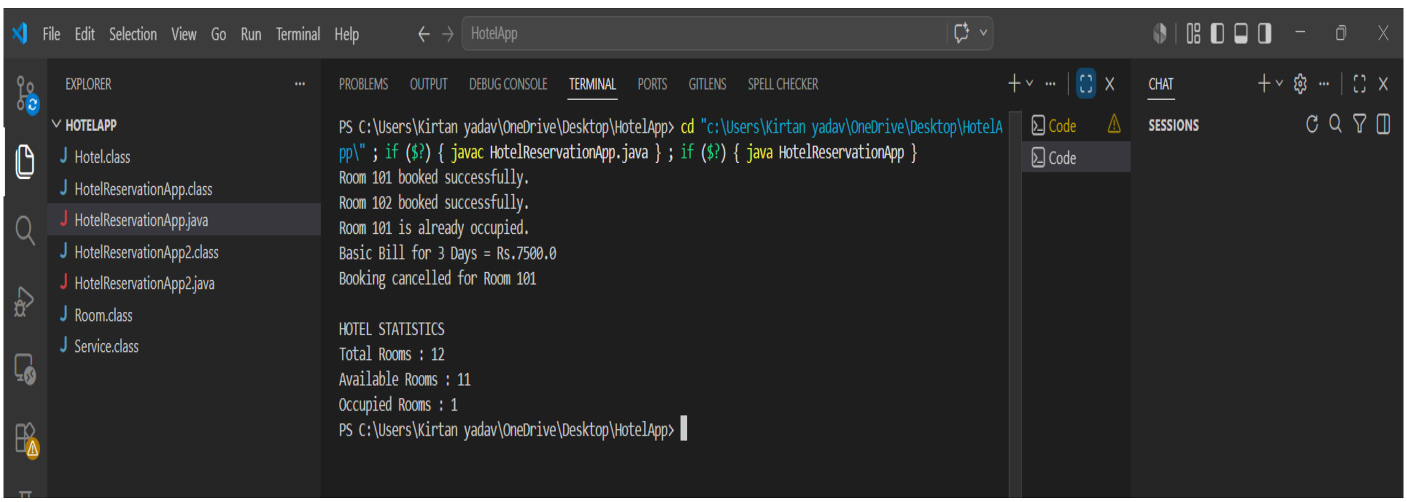
Implementation

The project consists of three major components: **1. Room Class:** The Room class stores room details such as room number, room type, price, and availability status. Encapsulation is implemented by making variables private and accessing them using getter methods. **2. Hotel Class:** The Hotel class manages all rooms using a 2D array. It handles booking, cancellation, room display, billing, and statistics generation. **3. Service Interface:** The Service interface provides abstraction by defining the billing method calculateBill(). The project successfully demonstrates interface implementation, real-world booking logic, array handling, and object state management.

Sample Input / Output

Output:

HOTEL ROOM RESERVATION SYSTEM



Gantt Chart

Task	Duration
Planning & Design	1 Day
Class Design	1 Day
Coding & Logic	2 Days
Testing	1 Day
Documentation	1 Day
Total Estimated Time	6 Days

Advantages

• Easy room management
• Automated billing system
• Reduces manual work
• Easy to maintain and upgrade
• Demonstrates real-world OOP implementation

Limitations

• Console-based application only
• No database support
• No online reservation feature
• No customer authentication system

Future Enhancements

• GUI implementation using Java Swing or JavaFX
• Database connectivity using MySQL
• Online booking integration
• Customer login system
• Payment gateway integration

Conclusion

The Hotel Room Reservation System project successfully demonstrates the practical implementation of Java and Object-Oriented Programming concepts in solving a real-world problem. The project helped in understanding classes and objects, encapsulation, interfaces and abstraction, 2D arrays, and booking and billing logic. The system provides an efficient method for managing hotel room reservations and billing operations. It also improves logical thinking, coding practices, and software development understanding. Overall, this project serves as a strong foundation for developing advanced hotel management systems in the future.

References

1. Java Programming – Herbert Schildt
2. Oracle Java Documentation
3. Object-Oriented Programming Concepts
4. VS Code Java Development Documentation